

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B. Tech (CBCS) Degree Examinations

Summer Semester July 2024
CV2105-1 – CONCRETE TECHNOLOGY
(CV)

Duration: 3 Hours

Max. Marks:100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions:

1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

2) Use of IS456-2000; IS10262-2019 permitted.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which cement contains high percentage of C3S and less percentage of C2S
A) Quick hardening cement
B) Expansive cement
C) Rapid hardening cement
D) Low heat cement
2. Silica in excess causes
A) Cement to set slowly
B) Cement to expand
C) Cement to disintegrate
D) Cement to set quickly
3. Which of the following is not a Bogue's compound
A) Tetra calcium aluminoferrite
B) Tricalcium aluminate
C) Tricalcium silicate
D) Tricalcium sulphate
4. Which compound may lead to a rapid stiffening of the paste with a large amount of the heat generation
A) C₂S
B) C₃A
C) C₄AF
D) C₃S
5. Lignosulphonic acids and their salts act as
A) super plasticizers
B) accelerators
C) set retarders
D) air entraining agents
6. Introducing wood resins into concrete act as
A) super plasticizers
B) set retarders
C) accelerators
D) air entraining agents
7. The use of air-entraining agents in concrete
A) increases bleeding
B) increases strength
C) increases workability
D) increases durability
8. Which of the following statement is incorrect about the function of admixtures
A) To retard the initial set, to keep concrete workable for a shorter time for placement
B) To reduce the segregation in grout and concrete mixers
C) To enhance the workability
D) To accelerate the initial set of concrete, to speed up the rate of development of strength at early ages
9. The measurement of workability by Vee Bee consistometer test is expressed in
A) minutes
B) seconds
C) centimeter
D) dimensionless
10. The ease with which the concrete can be mixed, placed and compacted in position in a homogeneous state is called.....
A) segregation
B) durability
C) workability
D) bleeding

11. Which of the following is not a method to determine the workability of concrete
 - A) compaction factor
 - B) flow test
 - C) vee-bee consistometer
 - D) soundness
12. Separation of cement paste from sand in the mortar allowing the water or cement paste to appear at the surface is called
 - A) bleeding
 - B) segregation
 - C) honeycombing
 - D) scaling
13. The modulus of rupture is a measure of
 - A) Direct tensile strength
 - B) Split tensile strength
 - C) Flexural tensile strength
 - D) Direct compressive strength
14. Concrete gains strength due to
 - A) Chemical action of cement with coarse aggregate
 - B) Hydration of cement
 - C) Evaporation of water
 - D) All of the above
15. Steam curing of concrete is adopted for
 - A) columns
 - B) Precast structures
 - C) Beams
 - D) walls
16. Schmidt's Rebound Hammer technique is used to measure
 - A) Shrinkage Limit
 - B) Thickness of member
 - C) Tensile Strength
 - D) Surface Hardness
17. has designated the concrete mixes into a number of grades as M10, M15
 - A) IS 456-2000
 - B) IS 456-2010
 - C) IS 513-1999
 - D) IS 465-2000
18. What is the approximate mix proportion for M10 grade concrete
 - A) 1:2:4
 - B) 1:1.5:3
 - C) 1:3:6
 - D) 1:1:2
19. Aerated Concrete is
 - A) Very heavy weight
 - B) Heavy weight
 - C) Medium weight
 - D) Light weight
20. The light-weight concrete is prepared by
 - A) Mixing Al in the concrete
 - B) Using coke-breeze, slag as aggregate in the concrete
 - C) Mixing Fe in the concrete
 - D) Mixing Portland cement with sawdust in specified proportion in the concrete

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the following types of cement with their usage in different situations: i).Rapid Hardening cement ii).Sulphate Resisting Cement iii).Portland Slag Cement iv).Super Sulphated Cement	8	L2	1	1
	b) Summarize the importance of mixing water quality and permissible impurity limits to be used in concreting as per BIS.	8	L2	2	1
2.	a) Write briefly about the following admixtures explaining its effect on concrete: i).Rice Husk Ash ii).Air Entraining Agents	8	L2	2	1
	b) Which type of cement is used for the following construction works and List the special properties of the same: i) Construction of Dam ii) Grouting the pre-stress concrete ducts.	8	L2	1	1

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| 3. a) | What is an admixture? What is the role of Water reducers, Air entraining agents and Accelerators in concrete. | 8 | L2 | 2 | 1 |
| b) | What is the role of Bogue's compound? Explain the heat of hydration with a schematic representation. | 8 | L2 | 1 | 1 |

Unit – II

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|-------|---|---|----|---|---|
| 4. a) | Explain with a sketch the method of measuring workability of fresh concrete by Slump Cone method and discuss the results. | 8 | L2 | 3 | 1 |
| b) | List any Six factors affecting durability of concrete. | 8 | L2 | 4 | 1 |
| 5. a) | Explain with a neat sketch the Ultra Sonic Pulse velocity method used for testing the hardened concrete. | 8 | L2 | 4 | 1 |
| b) | Define Segregation and Bleeding in concrete. List any two factors which affects segregation and bleeding in concrete. | 8 | L2 | 3 | 1 |
| 6. a) | Explain the following destructive methods of testing hardened concrete with relevant equations:
i).Compression strength ii).Split tensile strength | 8 | L2 | 4 | 1 |
| b) | Explain the following with respect to manufacture of concrete:
i).Transportation ii).Placing iii).Compaction iv).Curing | 8 | L2 | 3 | 1 |

Unit – III

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|-------|---|---|----|---|---|
| 7. a) | Explain any four types of natural and artificial aggregates used in the manufacture of light weight concrete. | 8 | L2 | 5 | 1 |
| b) | Define concrete mix design. Write the procedure of Mix Design with reference to the provisions laid in IS -10262- 2009. | 8 | L2 | 5 | 1 |
| 8. a) | Explain briefly the following types of concrete with respect to its materials, properties and applications:
i).High performance concrete ii).High density concrete | 8 | L2 | 5 | 1 |
| b) | Explain any four factors affecting the choice of mix design. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
